

Alcohol Intake and Blood Pressure: A Systematic Review Implementing a Mendelian Randomization Approach

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Evidence abstract

BACKGROUND Alcohol has been reported to be a common and modifiable risk factor for hypertension.

However, observational studies are subject to confounding by other behavioural and sociodemographic factors, while clinical trials are difficult to implement and have limited follow-up time.

Mendelian randomization can provide robust evidence on the nature of this association by use of a common polymorphism in aldehyde dehydrogenase 2 (ALDH2) as a surrogate for measuring alcohol consumption.

ALDH2 encodes a major enzyme involved in alcohol metabolism.

Individuals homozygous for the null variant (*2*2) experience adverse symptoms when drinking alcohol and consequently drink considerably less alcohol than wild-type homozygotes (*1*1) or heterozygotes.

We hypothesise that this polymorphism may influence the risk of hypertension by affecting alcohol drinking behaviour.

METHODS AND FINDINGS We carried out fixed effect meta-analyses of the ALDH2 genotype with blood pressure (five studies, n = 7,658) and hypertension (three studies, n = 4,219) using studies identified via systematic review.

In males, we obtained an overall odds ratio of 2.42 (95% confidence interval [CI] 1.66-3.55, $p = 4.8 \times 10^{-6}$) for hypertension comparing *1*1 with *2*2 homozygotes and an odds ratio of 1.72 (95% CI 1.17-2.52, $p = 0.006$) comparing heterozygotes (surrogate for moderate drinkers) with *2*2 homozygotes.

Systolic blood pressure was 7.44 mmHg (95% CI 5.39-9.49, $p = 1.1 \times 10^{-12}$) greater among *1*1 than among *2*2 homozygotes, and 4.24 mmHg (95% CI 2.18-6.31, $p = 0.00005$) greater among heterozygotes than among *2*2 homozygotes.

CONCLUSIONS These findings support the hypothesis that alcohol intake has a marked effect on blood pressure and the risk of hypertension.

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